Proton Teletherapy of Uveal Melanoma

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The optimal treatment of uveal melanoma would destroy the intraocular tumor, retain good vision, and avoid the development of metastatic disease. Unfortunately, inherent limitations make the development of such an optimal therapy unlikely in the near future.[1] One, approximately two thirds of choroidal melanomas examined in most ophthalmic oncologic practices are within 3 mm of the optic nerve or fovea. Even with a very focused type of treatment-delivery system, to avoid a marginal miss, usually such eyes have significant long-term visual damage. Two, whereas only 1% of patients present with simultaneously detectable metastases and a newly discovered uveal melanoma, the majority of patients who metastasize have already developed micrometastatic disease before the detection of their intraocular lesion. At present, there is no effective treatment of liver metastases from uveal melanomas. Three, in approximately 20% to 35% of patients, at the initial presentation, the tumor is too large for treatment with current eye salvage techniques, and these eyes are treated with primary enucleation.

In this issue, several different radiation delivery systems are presented, including those that rely on brachytherapy (radioactive plaques) as well as teletherapy (protons, intensity-modulated conformal therapy, and cyberknife). Over 10,000 patients with uveal melanoma have been treated around the world with charged particles (helium ions and protons).

There are 4 theoretical advantages of proton radiation. One, because of the inherent Bragg peak, the lateral and distal falloff with this form of treatment results in more focused radiation delivery than any other form of ionizing radiation. In most proton facilities, the lateral and distal falloff goes from 100% of doses to essentially zero in less than 3 mm. Two, there may be higher linear energy transfer with protons than with brachytherapy, especially the I-125 plaques. In a randomized, dynamically balanced, prospective study, with 70 Gy of I-125 plaque brachytherapy and helium radiation, there was significantly better tumor control with the latter modality.[2] Three, radiation delivery with protons does not subject the medical personnel to any radiation exposure, which is not possible in patients receiving brachytherapy. Four, with proton radiation, only a single surgical localization procedure is needed; with plaques, a radioactive plaque is inserted, and it must be removed.

The initial charged-particle radiation therapies for uveal melanomas were done in the United States with protons in Boston and with helium ions in Berkeley.[3,4] We initially chose to use helium ions because it had been assumed that melanomas were relatively radioresistant, and there is more linear energy transfer with heavier charged particles, which theoretically should be more efficacious. When it became apparent that there was no significant advantage with helium ions, and protons were more cost-effective, we switched to the latter form of treatment.

There are now several centers throughout the world that have had a large experience with proton radiation. There have been relatively similar results reported from those centers, with the caveat that there is a learning curve. Like both the Boston and San Francisco experience had seen many years ago, the Swiss group reported that their local control rate by decade had markedly improved from 91% in patients treated before 1988 to 96% in those treated between 1989 and 1993 to 99% after 1993.[5]

The indications for proton radiation of uveal melanomas vary in different centers.[6,7] In some countries, the logistics of proton radiation have limited their enthusiasm for its use. In other centers, because irradiated tumors within 3 mm of the optic nerve or fovea are likely to have a poor long-term visual outcome than treatment with other techniques, those alternative nonradiation therapies are often used first. In many ophthalmic oncology centers, because of a higher failure rate with brachytherapy for smaller melanomas in the posterior pole, especially those that surround or are adjacent to the optic disc, protons are used, whereas more anterior tumors can be treated with any radiation form with almost equivalent local tumor control.[7,8]

We use proton radiation to treat most medium and large uveal melanomas. There are several exceptions as well as contraindications to this therapy. In patients with thin, growing (<3.5 mm) melanomas that are less than 10 mm in diameter that are either adjacent to the nasal disc or within 3 mm of the fovea, we prefer to attempt to destroy them with laser-induced hyperthermia. Our choice of that modality is predicated on the long-term visual loss associated with proton radiation of such lesions, and the fact that although there is a significantly higher failure rate with laser, often we are able to keep excellent long-term vision.[9] If such patients are closely followed, we have not had any tumor-related mortality in those that have developed recurrent growth that required radiation (unpublished data). In very thick (>9 mm), choroidal or ciliochoroidal melanomas that are less than 15 mm in diameter, several groups, including ourselves, have found that there appears to be less ocular morbidity with eyewall resection followed by adjunctive radiation than with either radioactive plaque or proton radiation alone.[10–12] We have excluded patients from proton treatment in which over 40% of the eye volume is involved by tumor, patients who have neovascular glaucoma before treatment, when there is a large, extraocular, localized orbital extension involving the orbit, or eyes that are painful and blind. In those latter settings, we have not been able to salvage a useful eye.

To treat patients with proton radiation, most surgeons routinely place 4 tantalum marker rings on the scleral surface surrounding the lesion. This is shown schematically in Figure 1. Often at the time of treatment, especially if there is uncertainty about the diagnosis, a fine needle aspiration biopsy (FNAB) is performed. We have found essentially no morbidity with this technique. We have seen no evidence of tumor spread, and this data adds additional prognostic information.[13] We are currently using new molecular techniques on FNAB samples to further improve the prognostic accuracy in such cases.

In most centers, patients are treated between 2 days and 2 weeks later with 5 fractions of proton radiation. Our group and the group in Boston have performed dose studies. We found no significant difference in control rate, complications, enucleation rates, or metastases among 50, 60, 70, and 80 GyE of helium ion irradiation.[14] The Boston group similarly has noted no difference between 50 and 70 GyE with proton radiation.[15]

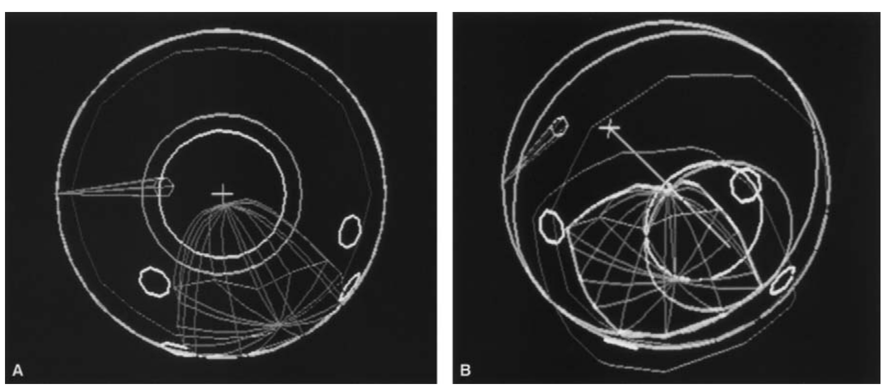

Figure 1. *(**A** and **B**) Schematic of tantalum marker ring placement (circles) on the scleral surface surrounding the lesion, radiation field (blue line), tumor (red), fovea (pink +), and optic disc (pink circle). Alteration of beam and gaze position from neutral (**A**) to down and lateral (**B**) increases distance from high-dose radiation to both optic nerve and fovea. Reproduced from Char DH. Tumors of the Eye and Ocular Adnexa.* BC Decker; 2001:162.

In our experience with brachytherapy and both proton and helium ion teletherapy, there are slightly more rapid regression rates in patients who are treated with high-dose brachytherapy.[2] Usually after proton radiation, tumors either shrink very slowly or initially are stabilized. In less than 5% of successfully treated patients, there is a small, transient increase in thickness. Usually there is a mean latency of approximately 6 months before the first detection of radiation response, which is loss of subretinal fluid. The mean latency between treatment and initial detection of tumor shrinkage is often delayed for approximately 1 year. Several authors have published data with both particles and brachytherapy that showed that more rapid melanoma shrinkage is associated with adverse prognosis. Although this may appear to be contraintuitive, many studies have shown that more rapidly cycling tumors are associated with a worse prognosis. More rapidly cycling tumors, when irradiated, will shrink more rapidly because more cells are going into mitoses, and the DNA-induced errors from radiation become manifest as cell destruction.

Tumor shrinkage is a delayed process that can continue for as long as 5 years (Fig. 2). In approximately 10% of patients with successfully irradiated uveal melanoma, there is no shrinkage, but the tumor remains stationary. In our experience, such eyes removed for late complications or obtained at autopsy have no cells that are reproductively intact, as judged by the bromodeoxyuridine, Ki-67, or other labeling techniques. In most series, there is approximately a 40% mean shrinkage in tumor volume after successful radiation.

In the experience from San Francisco, Boston, and Switzerland using current techniques, there is >98% local control rate. Long-term retention of eyes is dependent on tumor size and thickness. In a recent Boston report, approximately 84% of eyes were retained at 15 years.[3–5]

The major advantage of charged-particle radiation has been the very high local tumor control rate. No brachytherapy reports with similar uveal

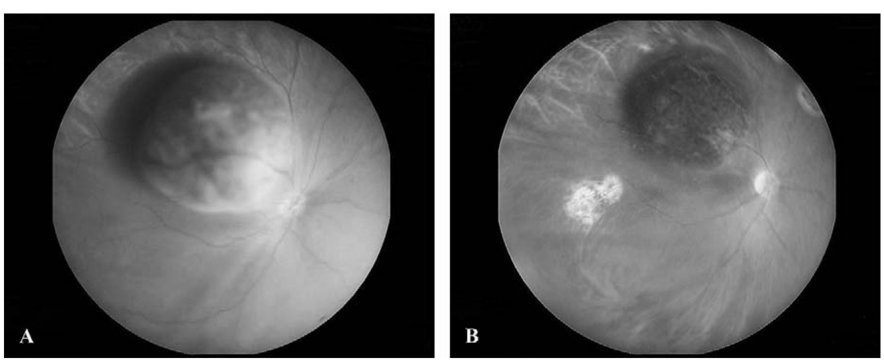

Figure 2. *(**A** and **B**) Preproton radiation choroidal melanoma (**A**). Posttreatment (**B**), the first sign of radiation response is loss of subretinal fluid with a mean latency of 6 months. Usually tumor size regression is delayed for 1 year and can continue over 5 years.*

melanomas and long-term follow up have shown equal control. The early experience, especially from our randomized, prospective, dynamically balanced study, showed significant side effects with protons, especially eyelash damage and neovascular glaucoma, which were not observed as frequently with radioactive plaques.[2]

Some radiation complications are preventable, whereas others are a function of tumor size and occur regardless of the type of radiation used. As an example, in a group of large uveal melanomas treated with iodine brachytherapy (mean diameter of 16.1 mm and mean thickness of 10.7 mm) with 3.5 years of follow up, 69% developed cataract and 60% developed glaucoma.[15] Neovascular glaucoma has historically been the most deleterious side effect of proton radiation.[16] Unlike neovascular glaucomas of other types, often these patients will not have obvious new vessels, except visible on gonioscopy of the iris–cornea angle. We observed over 30% of patients treated with helium ions developed neovascular glaucoma. Many of these were very large tumors, but at the time we initially started that treatment trial, we attempted to avoid optic nerve and macular radiation at the expense of inclusion of other structures in the treatment beam. In a retrospective study, we noted that the percent of the anterior segment in the entrance beam is highly correlative with development of neovascular glaucoma (Fig. 3).[17] Techniques to avoid the anterior segment, such as the use of multiple overlapping beams (Figure 4), have decreased our neovascular glaucoma rate to under 8%.

Neovascular glaucoma (NVG) development after radiation is multifactorial. In patients with diabetes, there is a synergistic effect between the diabetic retinopathy and the radiation vasculopathy, and such patients are at greater risk for developing NVG.[19] As we published, panretinal photocoagulation is useful treatment of neovascular glaucoma if the angle is still open.[16] There is also an inflammatory component, and in some patients, treatment with topical Pred Forte and cycloplegia has been effective.

If a large uveal melanoma is in the anterior part of the eye so that the eyelashes are irradiated either in the entrance beam with proton teletherapy or from the radiation scatter from brachytherapy, eyelash loss occurs.

The development of radiation optic neuropathy and maculopathy depends on whether each of these structures is in the high-dose region of the particle beam. Over 80% of patients with choroidal melanomas <3 mm from the optic disc and/or fovea have poor vision 5 years after proton radiation.[2–5] Visual results are predicated on several independent pretreatment parameters: tumor location vis-à-vis the optic nerve and macula, tumor thickness, and patient age. In tumors that are <6 mm in thickness and >3 mm from the optic nerve and fovea, approximately 75% will retain better than 20/40 vision after treatment.[2] Unfortunately, in the converse setting, either with tumors >6 m thick or thinner tumors <3 mm from the optic nerve or the fovea, over 80% will not retain good vision 5 or more years after proton radiation.[2] What is puzzling is a subset of patients with

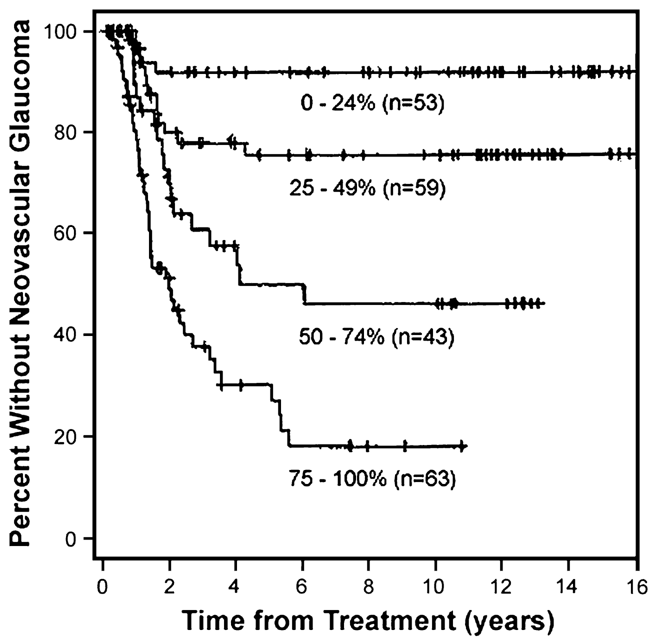

Figure 3. *The percent of the anterior segment in the entrance beam is highly correlative with development of neovascular glaucoma after charged-particle radiation. Kaplan-Meier univariate estimates of the percent of patients with neovascular glaucoma stratified by the percent of lens in port. Hash marks correspond to censored observations. Reproduced from Meecham WJ, Char DH, Kroll S, et al. Anterior segment complications after helium ion radiation therapy for uveal melanoma. Radiation cataract. Arch Ophthalmol.* 1994;112:197–203.

tumors that do not involve the macula but in which the disc received a full radiation dose; in 40% of the patients who were relatively young with thin tumors, good long-term vision has been retained (unpublished data).

Many patients with peripheral choroidal melanomas, because of the sharp edge of the proton beam, will develop an off-axis lens change (usually posterior subcapsular and anterior subcapsular change) and not develop a visually significant cataract. If >50% of the lens is in the entrance beam, a visually significant cataract usually occurs.[17] Unlike neovascular glaucoma, which either has developed within 3 years of treatment or does not occur, the development of lens opacities is a continuing effect of radiation.[17]

There are multiple etiologies for tumor recurrence after proton radiation therapy. One, as discussed previously, there is a learning curve in centers just beginning treatment.[5] Two, there are some cases in which the boundaries of the tumor histologically extend far outside those apparent on clinical or ultrasound examination. Several cases of diffuse uveal melanomas that initially appeared focal and therefore were not entirely treated have been reported.[20] Three, in some very large tumors, it is

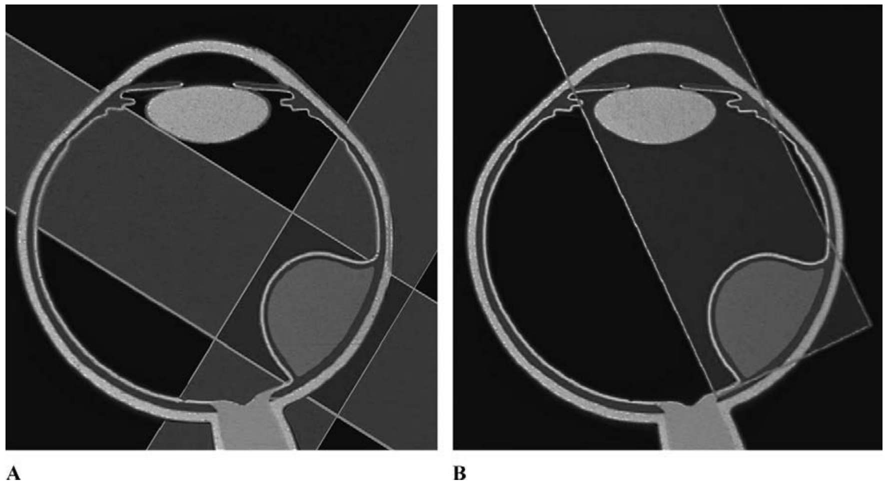

Figure 4. *(**A** and **B**) The use of multiple overlapping beams (**A**), in contrast to single beam (**B**), decreases anterior segment radiation dose and diminishes risk of neovascular glaucoma.*

possible that there has been sufficient tumor necrosis to produce vascular insufficiency to make them radioresistant. Almost all the failures with protons have occurred within 3 years of treatment. In a retrospective study, we studied late failures (>5 years after therapy), and we had only one case now develop tumor failure at 5 or more years after proton or helium ion radiation. There is a similar paucity of late proton failures in the Boston and Swiss experience. In contradistinction, approximately 2% of patients per year after 1-125 brachytherapy have late failures.[21] That latter data mirror the brachytherapy results seen with prostate treatment.

Whereas local intraocular uveal melanoma control with proton teletherapy exceeds 98%, tumor-related mortality, as discussed at the outset of this chapter, is predicated on the same parameters noted in primary enucleation eyes.[22] Patients with large-diameter tumors, more malignant cell types, older patients, and tumors that crossed the equator have a higher incidence of metastatic disease. Overall, in most charged-particle radiation series, less than 20% of patients with uveal melanoma have developed metastatic disease. The experience in all the proton treatment centers has suggested that survival in patients treated with radiation is no worse than those who were treated with enucleation, as noted with plaques versus enucleation in the COMS study.

The major challenges with proton radiation are to decrease both treatment-related morbidity and tumor-related mortality. Unfortunately, as discussed at the outset of this chapter, both of these challenges are unlikely to be met unless newer adjunctive therapeutic approaches can be developed. Most patients who develop metastatic disease already have micrometastases before treatment, and unfortunately, even with a highly focused charged particle, many tumors are in such close proximity to visually vital structures that the visual outcome in such cases at 5 years is poor.

It is uncertain if there are unknown negative radiation effects on uveal melanoma patient survival. Historically, we have assumed that the radiation-induced DNA-induced errors result in melanoma cell death and this decreased tumor-related mortality. Whereas this is most likely the case, there are other possible radiation–tumor interactions. In the D-12 model, single high-fraction radiation of a primary cutaneous mouse melanoma decreased antiangiogenic factors produced by the tumor, which results in the rapid growth of dormant micrometastases.[23] In another model, low-dose radiation upregulated the nitric oxide pathway with increased angiogenesis.[24] In animal sarcoma models, particle radiation appears to be more effective at suppressing metastatic disease than photon radiation.[25]

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